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Practice Assignment • 30 min

English Due Oct 25, 11:59 PM IST English S

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Week 5: Image Stitching

- **Ungraded Plugin:** 5.1 Overview of Image Stitching
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- **Ungraded Plugin:** 5.2 2x2 Image Transformations
  9 min
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- **Ungraded Plugin:** 5.3 3x3 Image Transformations
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Week 5 Quiz

Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

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5.3 3x3 Image Transformations Self-check Quiz

coach

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice Let's chat

Assignment details

Due
Oct 25, 11:59 PM IST

Attempts
Unlimited

Submitted
Oct 20, 3:01 AM IST

Try again

View submission See feedback

1. The point (4,2,2) is equivalent to which of the following points in homogeneous coordinates? **1 / 1 point**

○ (4,2,4)
○ (4,2,1)
● (2,1,1)
○ None of the above

Correct
(x, y, z) represents a point at location $(x/z, y/z, 1)$. From this, it follows that $x = 2$ and $y = 1$ for the given homogeneous coordinate.

2. Which operation does the following 3x3 matrix represent? **1 / 1 point**

$$\begin{bmatrix} \tilde{x}_2 \\ \tilde{y}_2 \\ \tilde{z}_2 \end{bmatrix} = \begin{bmatrix} 1 & m_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ 1 \end{bmatrix}$$

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○ Translation
○ Rotation
● Skew
○ Scaling

Correct
Please refer to the lecture video

3. Imaging a plane through a pinhole is an example of which transformation? **1 / 1 point**

○ Affine
● Projective

Correct
A pinhole camera maps a plane in the scene to a plane in the image plane through the pinhole. This mapping is a projective transformation that can be represented by a homography.